

Taxonomy

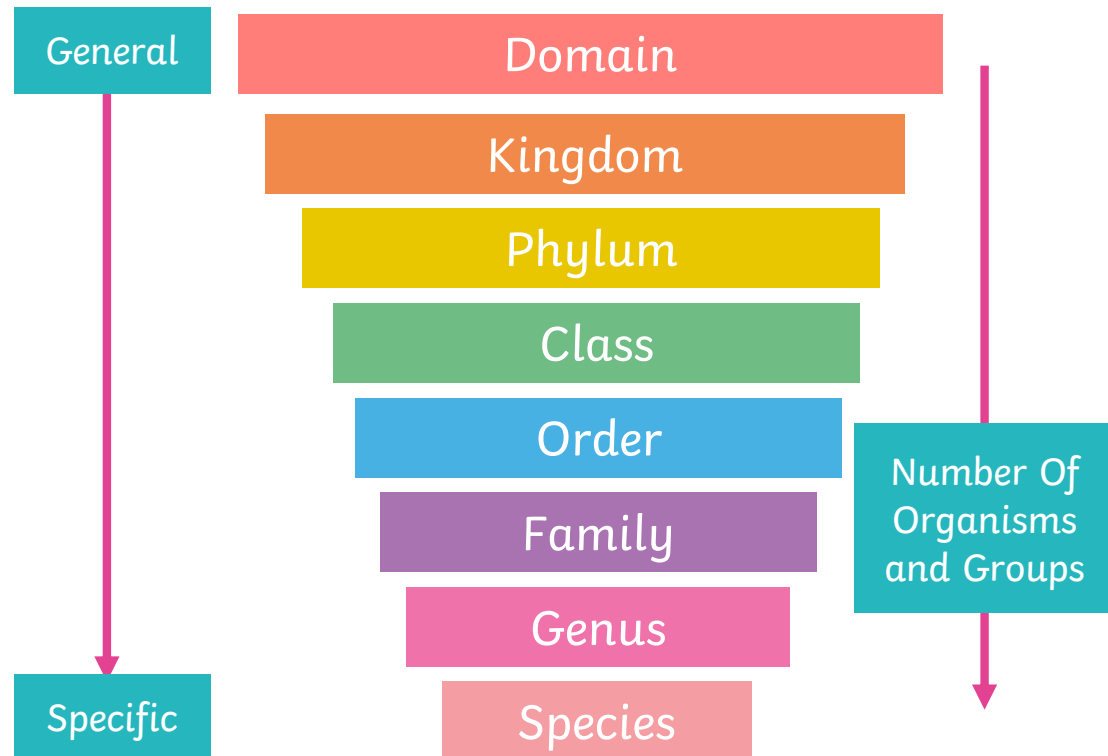
Sorting the World



What is Taxonomy?

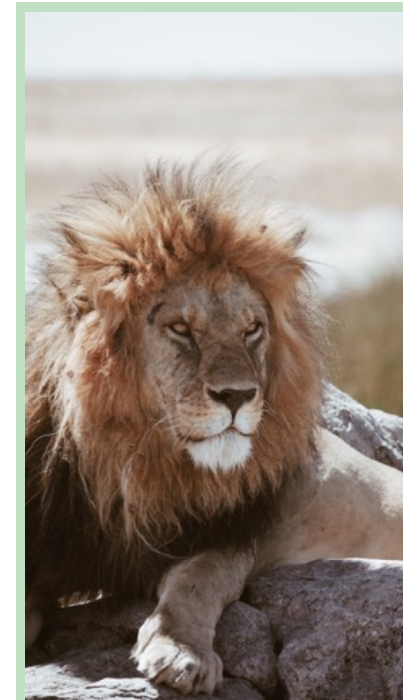
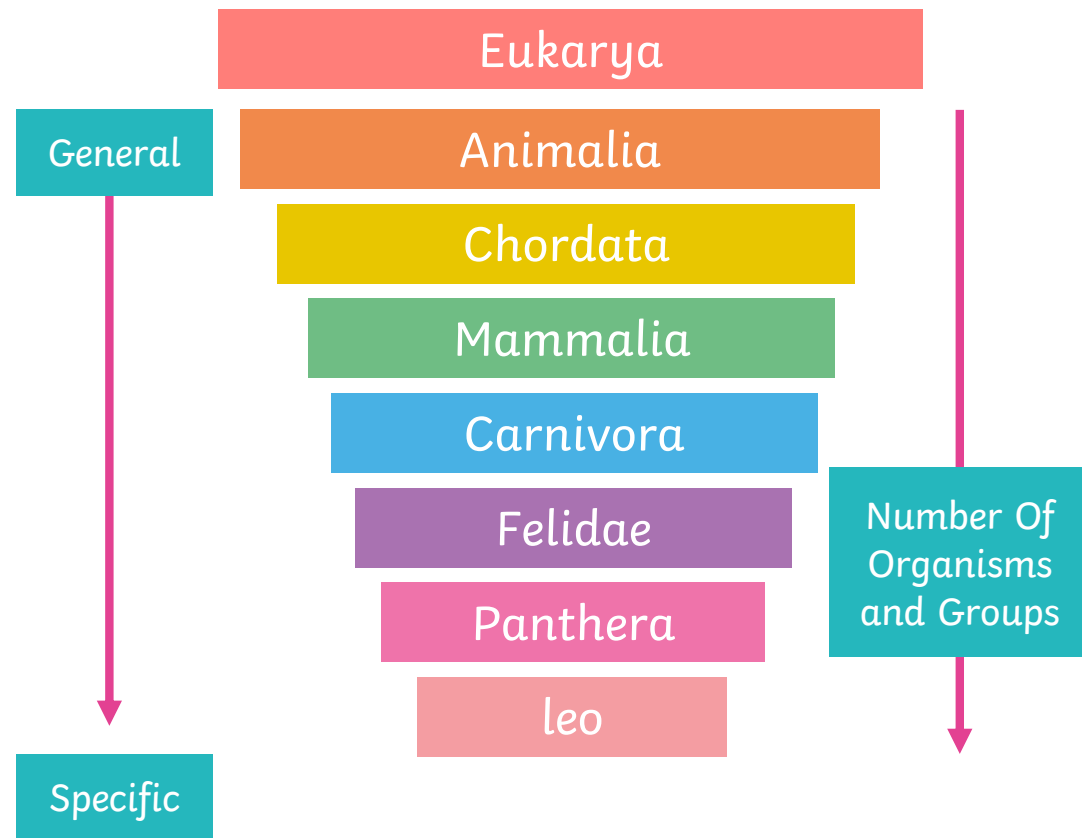
Taxonomy = classifying and naming things.
In biology, taxonomy is the classifying and naming of living organisms into related groups. In other words, sorting the world!

Taxonomy is a one stop shop for all the information you need to know about a living organism divided into **taxonomic categories**. It is a **hierarchical** system which starts at the broadest category, with the highest numbers. The categories become more precise and the numbers in each group drop as the levels progress.



How does it work?

Here's an example of the taxonomic classification of a lion.



The Scientific Name

To keep it simple, Linnaeus had the idea to give each organism a two-part Latin scientific name made up of the **genus** and **species** - the last two parts of taxonomic categories. He wanted to make sure that no two living organisms had the same name, so there was no confusion.

The scientific name of a lion is always:

Lion = Panthera leo

Species

Genus - always with a capital letter.



Why use Latin? Discuss with a buddy.

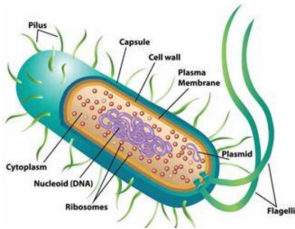
In Linnaeus's time, Latin was considered the language of scholars but was also a dead language. No country used Latin as their official language which made it perfect for his naming system. The Latin name for any living organism would be the same no matter where you were in the world.

****Keep your eyes and ears open! You'll begin to notice that many of our modern English words are derived from the Latin language.**

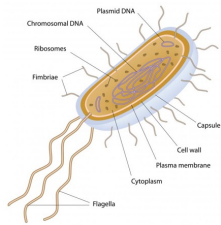
Domain

The idea of domains started when scientists were able to study the genes of living things. They found that many single-celled organisms that had been grouped together were only distantly related. In 1990, Carl Woese proposed that “Domain” could be used as the highest rank, based on the cell structure of living things.

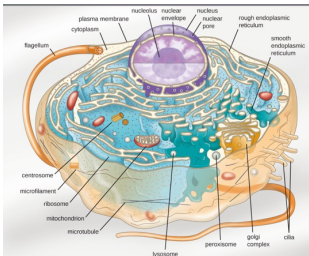
In this classification system, there are 3 domains



Bacteria (Eubacteria) – single-celled organisms whose cells have no nucleus. They are found everywhere on earth and form both symbiotic and parasitic relationships with humans



Archaea (Archaeobacteria) – single-celled organisms whose cell chemistry is different than other bacteria. They can be found in extreme environments like thermal vents and places where there is no oxygen or high acidity.



Eukarya (Eukaryota) – Includes all living things whose cells have a nucleus. All animals, plants, fungi and many single-celled organisms are eukaryotes.



Kingdom

Once considered the broadest or highest level of classification, as we learn more about living things, scientific ideas about kingdoms are debated and can change. In modern classification, all members of a kingdom should share a common ancestor.

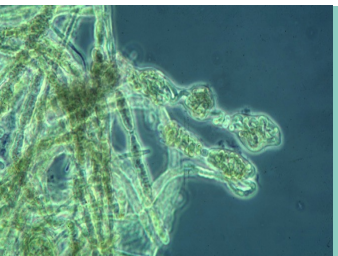
Most scientists now agree on 7 kingdoms. These are:



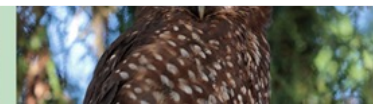
Bacteria and Archaea (same as the Domains!) These two groups are also their own kingdoms. One difference is that some bacteria can go dormant for a long time (called an Endospore) while no known archaea can.



Protozoa (Single celled eukaryotes) – These microorganisms are sometimes called “one-celled animals” because they move and eat like animals and do not have a cell wall



Chromista (eukaryotes that can photosynthesize) - They include single-celled and multicellular organisms, such as algae. More research is needed to know how closely different species in chromista are related



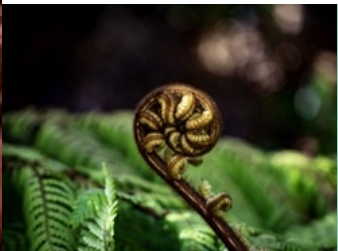
Kingdom



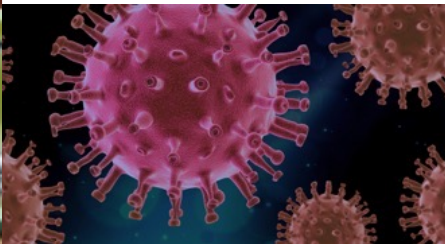
Animalia (Animals) - Over 1 million known species from insects and worms to whales and humans.



Fungi (Multi cellular organisms) - They include things like mushrooms, mold and mildew.



Plantae (Plants) - Over 380,000 known species, from mosses to large trees.



Some scientists believe there should be a kingdom for viruses. What do you think?



Phylum

Each kingdom is divided into phyla. **Phyla** is the plural term for **phylum**. These are less broad than a kingdom but are organised by general characteristics.

In the animal kingdom, there are more than 30 phyla. Too many to list here!

Here are a couple of examples:



Phylum Chordata — This includes all living organisms that have a backbone or spinal cord. Humans are included in this phylum.



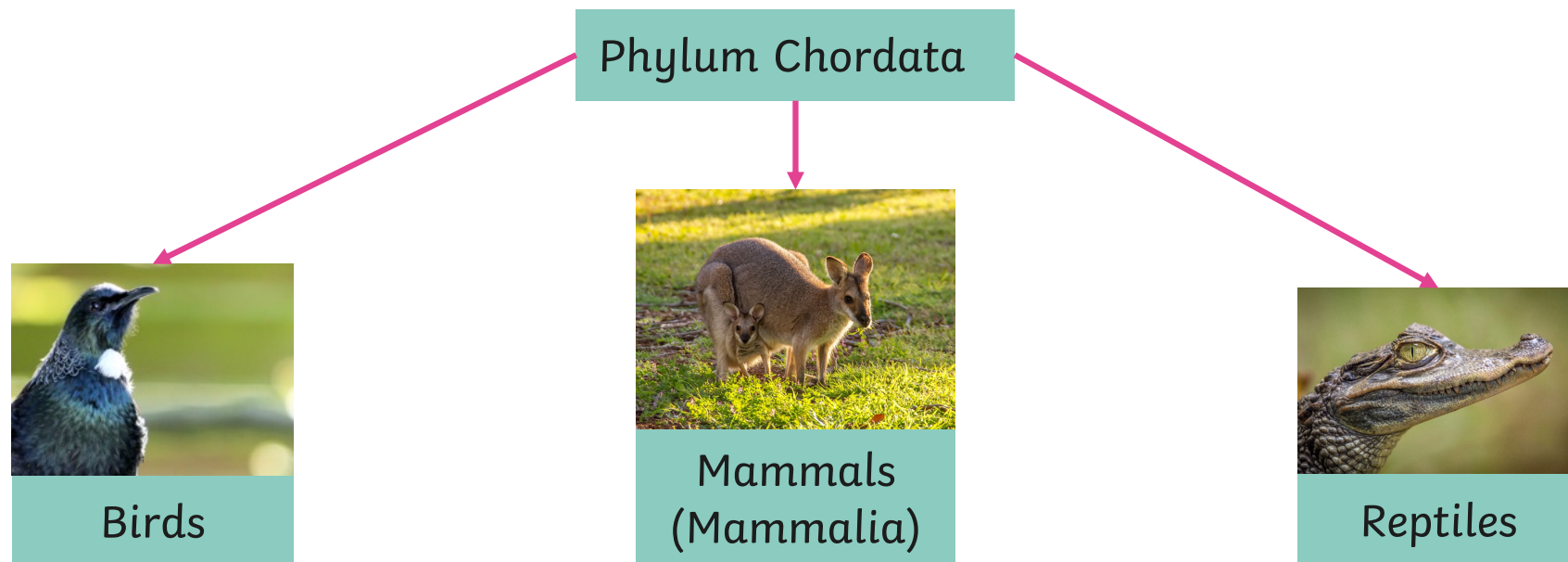
Phylum Arthropoda - This phylum includes insects, spiders and other animals with segmented bodies, such as worms, ants and flies. Arthropods have their skeletons on the outside of their bodies e.g. a crayfish.



Class

Each phylum is divided further into classes. This further refines similarities within the groups.

For example:



Question: What are the characteristics of a mammal?

Order

Each class is divided into smaller groups called orders. There is some disagreement amongst scientists at this level. Orders may differ depending on the scientific source.

Here's some examples:



Lepidoptera is the order of moths and butterflies.



Carnivora is the order within Mammalia that is distinguished by having powerful jaws and teeth that can stab, tear and eat flesh. They include animals like cats, dogs, bears, hyenas, racoons and wolves.

Question: Are humans carnivores?

Family

By this stage of the hierarchy, the different characteristics of living organisms are becoming more specific, so there are more family groups to choose from. Consequently, there is disagreement amongst scientists about which family an animal or plant might belong to. Sources will vary, so you often need to find the best consensus from a range of opinions.

Here are some examples of family groups:



Dogs = Canidae



Cats = Felidae



Owls = Strigidae



Frogs = Ranidae



Humans = Hominidae



Bees = Apidae

Genus

The **genus** may only have one or two animals in it. If they have the same **genus** this means they are closely related.

For example the **genus** of these two animals:



Dog



Wolf

Canis



Species



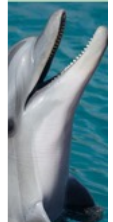
The **species** is the most basic level of taxonomy. A **genus** can be made up of many different species.

Species are groups of organisms that have:

- similar features that are unique to them
- similar form and structure
- similar habitats

Species can also have **subspecies**. We know so much more about the world around us than Carl Linnaeus did in the 18th century that new biological discoveries are being made all the time. Consequently, new subspecies need to be classified and named.

This means that sometimes a living organism may have a three part name to show it has slightly different characteristics to its relative.



Species

For example here are three of the nine subspecies of the leopard.



Panthera pardus pardus
(African Leopard)



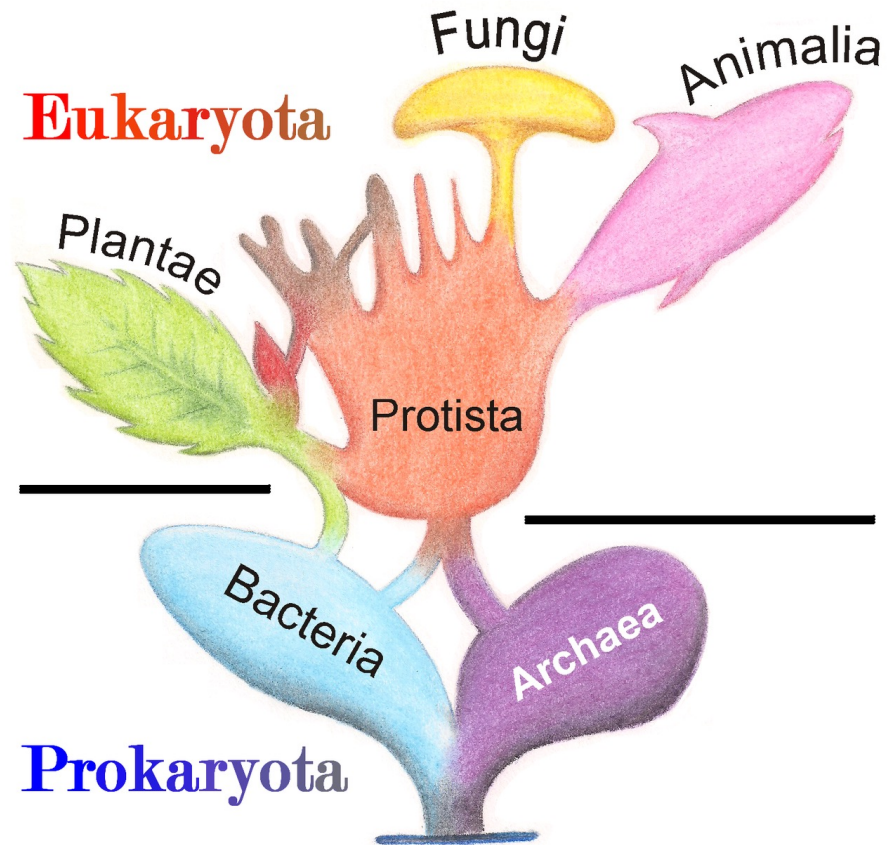
Panthera pardus saxicolor
(Persian Leopard)



Panthera pardus orientalis
(Amur Leopard)

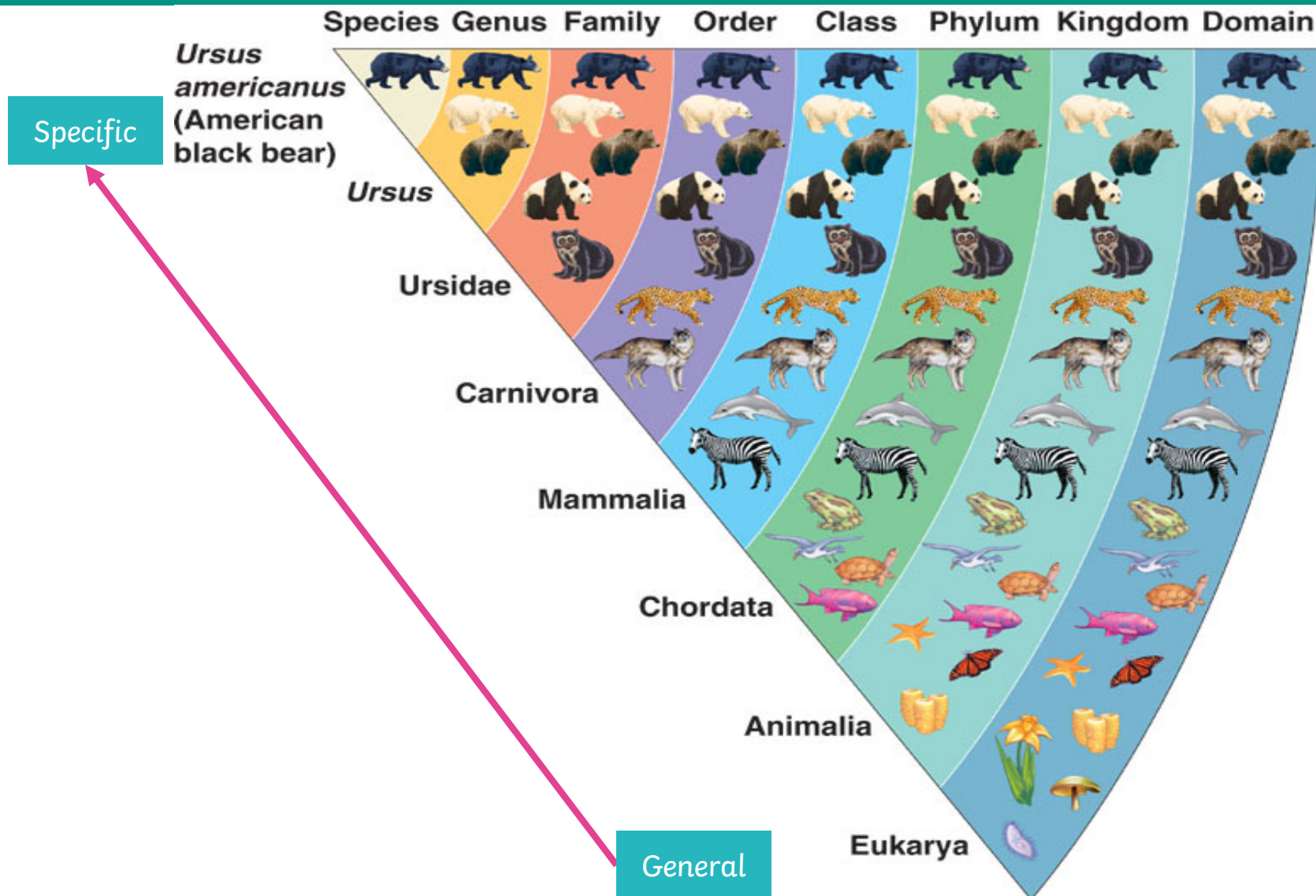
"African leopard, *Panthera pardus pardus*, near Lake Panic, Kruger National Park, South Africa" by Derek Keats is licensed under [CC BY 2.0](https://creativecommons.org/licenses/by/2.0/).

The Tree of Life



All living things are related! Some organisms are more closely related than others.

Example: Black Bear



Can You Make a Better Mnemonic?

Mnemonic Device

**“Dear King Phillip came over for
good spaghetti!”**

- Domain, Kingdom, Phylum,
Class, Orders, Family,
Genus, Species.

